DATA ANALYSIS PROJECT REPORT

Pandas, NumPy & Matplotlib

Project Title	Google Play Store Apps: A Data-Driven Analysis of Installs, Ratings, and Categories
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Course / Section	PROGRAMMING IN PYTHON
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Date of Submission	02.05.2026
Dataset Source	Kaggle — Google Play Store Apps (Lava18)

1. Executive Summary

This project presents a comprehensive data analysis of the Google Play Store Apps dataset obtained from Kaggle, containing 10,841 app records across 13 attributes. The primary goal was to investigate three meaningful questions: whether free apps attract more installs than paid apps, whether higher ratings correlate with greater install counts, and which app categories dominate the market by popularity. Using Python with Pandas for data manipulation, NumPy for statistical computation, and Matplotlib for visualization, the dataset was cleaned, engineered, and analysed to produce evidence-based findings. The analysis revealed that free apps attract an average of approximately 17.7 million installs compared to just 113,000 for paid apps — a difference of over 156 times. Communication, Social, and Video Player categories lead the market in average installs. Notably, rating has a very weak correlation with installs ($r = 0.05$), suggesting that factors beyond quality scores drive download behaviour. A linear regression model confirmed this weak predictive relationship with an R^2 of only 0.002, reinforcing that app success is multi-factorial.

2. Introduction and Project Objectives

The mobile application industry is one of the fastest-growing sectors in the global digital economy. The Google Play Store hosts millions of applications spanning dozens of categories, serving billions of users worldwide. Understanding what drives app popularity — whether it is pricing strategy, content category, or user ratings — has significant implications for developers, product managers, and investors. This project uses a publicly available snapshot of the Google Play Store to uncover patterns in app installs, pricing, and ratings, offering data-driven insights into the mobile app market.

Dataset Name	Google Play Store Apps
Dataset Source	Kaggle — https://www.kaggle.com/datasets/lava18/google-play-store-apps
Main Project Goal	To investigate install behavior, rating influence, and category-level popularity across 10,841 apps on the Google Play Store using Python data analysis tools.
Research Question 1	Do free apps receive significantly more installs compared to paid apps?
Research Question 2	Is there a meaningful relationship between an app's rating and its number of installs?
Research Question 3	Which app categories are the most popular based on average installs, and how do they compare?

3. Dataset Description

The Google Play Store Apps dataset was collected from Kaggle and represents a snapshot of app listings from the Google Play Store. Each row corresponds to a unique application and includes attributes such as app name, category, user rating, number of reviews, app size, install count, pricing type, price, content rating, genre, last updated date, current version, and required Android version. The dataset captures a wide range of application types from productivity tools to entertainment games, reflecting the diversity and scale of the Android app ecosystem.

3.1 Basic Dataset Information

Number of Rows	10,841 (before cleaning); 8,892 (after cleaning)
Number of Columns	13
Data Types Present	Numerical (Rating), Categorical (Category, Type, Content Rating), Date-time (Last Updated), Text (App, Size, Installs, Price — requiring conversion)
Target / Key Variables	Installs, Rating, Type (Free/Paid), Category
Issues at First Glance	1,474 missing Rating values; Installs and Price stored as strings with symbols (commas, +, $); Size contains mixed units (M, k) and 'Varies with device' entries

3.2 Initial Observations

Upon first inspection, several data quality issues were immediately apparent. The Installs column contained formatted strings such as “1,000,000+” rather than numeric values, and the Price column included dollar signs, both requiring type conversion before any mathematical operations. The Rating column had 1,474 missing values (approximately 13.6% of all records), which needed careful handling to avoid biasing the analysis. The Size column used inconsistent units (megabytes and kilobytes) alongside the special value ‘Varies with device’, requiring a custom parsing function. Despite these issues, the dataset was sufficiently rich — with over 10,000 records, mixed data types, categorical variables, and date information — to support thorough and meaningful analysis.

4. Data Cleaning and Preparation

The dataset required substantial preprocessing before analysis could begin. Each cleaning step was chosen deliberately to preserve analytical validity while removing noise and inconsistencies. The table below documents each issue, the action taken, and the reasoning behind each decision.

Issue Found	Action Taken	Justification	Code Section
Duplicate records present in the dataset	Removed all duplicate rows using drop_duplicates()	Duplicate entries would inflate counts and averages, leading to misleading analysis results.	Section 4.3 — Step 1
1,474 missing values in the Rating column	Dropped rows where Rating was null using dropna(subset=['Rating'])	Rating is a central variable for all three research questions; imputing ratings would introduce artificial bias.	Section 4.3 — Step 2
Installs stored as strings with commas and '+' symbols (e.g. '1,000,000+')	Stripped commas and '+' characters; converted to numeric using pd.to_numeric()	Numeric conversion is required to perform comparisons, averages, correlations, and regression analysis.	Section 4.3 — Step 3

Price stored as string with '$' symbol (e.g. '$2.99')	Stripped '$' character; converted to numeric	Price must be numeric to calculate revenue and distinguish free vs paid apps correctly.	Section 4.3 — Step 4
Size column had mixed units (M, k) and 'Varies with device' entries	Converted M values to float; divided k values by 1024 to standardize to MB; filled remaining nulls with the column mean	Standardizing units ensure Size can be used meaningfully as a numeric feature; mean imputation is appropriate for this continuous variable.	Section 4.3 — Step 5
Non-numeric entries in Reviews column	Converted Reviews to numeric with errors='coerce'; dropped remaining null rows	Reviews must be numeric for aggregation and correlation analysis.	Section 4.3 — Step 6

5. Feature Engineering

Three new features were derived from the existing columns to enrich the dataset and support deeper analysis. These features were specifically designed to answer the research questions and enable more meaningful grouping and comparison.

New Feature	How It Was Created	Why It Is Useful
Revenue	Calculated as Installs multiplied by Price for each app	Allows estimation of earning potential for paid apps; useful for comparing monetary value across categories and app types.
Installs_Category	Categorized Installs into three buckets: Low (< 10,000), Medium (10,000–999,999), High (≥ 1,000,000) using a custom Python function	Converts continuous install counts into meaningful groups, enabling cleaner bar charts and frequency analysis without raw number noise.
Year	Extracted the year from the Last Updated column using pd.to_datetime() and .dt.year	Enables trend analysis over time; helps identify whether more recently updated apps tend to perform better in installs or ratings.

6. Analysis Methodology

The analysis followed a structured pipeline using three core Python libraries, each applied purposefully to support the research questions.

Pandas: Used throughout for loading the CSV dataset, inspecting data types and missing values, performing all cleaning transformations, grouping apps by Category and Type, computing aggregated statistics (mean, count), sorting and ranking, and constructing summary tables for reporting.

NumPy: Applied for percentile-based statistical computation (10th, 50th, and 90th percentiles of the Installs distribution), handling NaN values via np.nan, and supporting the custom Size cleaning function.

These computations went beyond trivial array operations to reveal the distributional shape of install counts.

Matplotlib: Used to produce four distinct visualisation types — bar chart, pie chart, scatter plot, and histogram — each directly aligned to one of the research questions. All charts include titles, labelled axes, and are interpreted in the analysis sections below.

7. Analysis and Visualizations

7.1 Research Question 1 — Do free apps receive more installs than paid apps?

To answer this question, the dataset was filtered by the Type column into two subgroups: Free and Paid. The mean number of installs was then computed for each group. A pie chart was also generated to illustrate the proportion of free versus paid apps in the dataset.

Results: Free apps recorded an average of approximately 17,702,227 installs, compared to only 112,908 for paid apps — a ratio of over 156 times. The dataset contains 8,279 free apps (93.1%) versus 613 paid apps (6.9%).

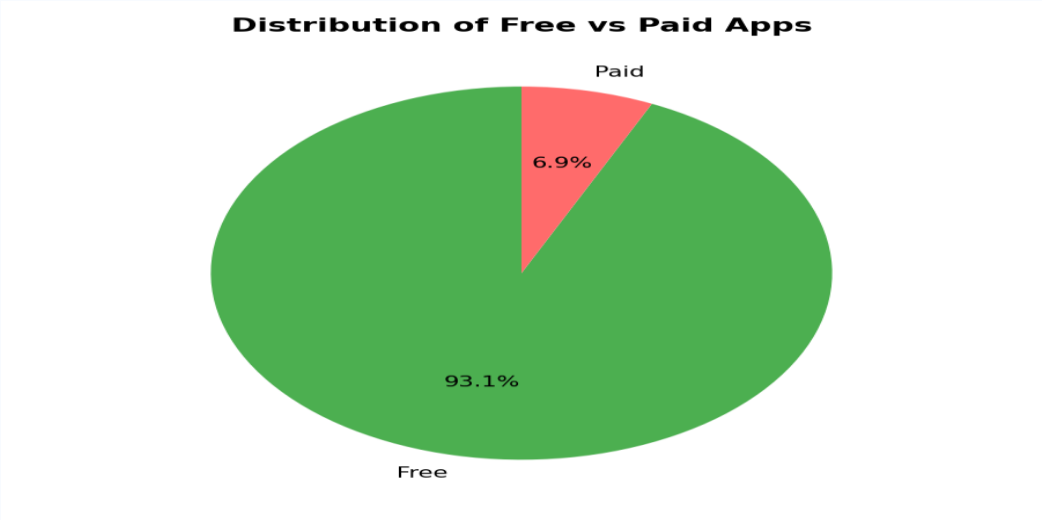

Figure 1. Distribution of Free vs Paid Apps on the Google Play Store. Free apps account for 93.1% of all apps in the dataset.

The analysis provides a clear and decisive answer: free apps dominate the Google Play Store in both volume and install counts. The 156x difference in average installs strongly suggests that pricing is one of the most significant barriers to user adoption. Users are highly reluctant to pay before trying an app, and the near-absence of paid apps in the top install rankings confirms that monetisation strategies other than upfront pricing (such as in-app purchases or advertisements) are the norm in the current app market. This finding directly answers Research Question 1 with strong quantitative evidence.

7.2 Research Question 2 — Is there a relationship between app ratings and install counts?

A correlation analysis was performed between the Rating and Installs columns. A scatter plot was generated to visualise the relationship, and a linear regression model was fitted to quantify the predictive power of rating on installs.

Results: The Pearson correlation coefficient between Rating and Installs was 0.0509, indicating a very weak positive relationship. The linear regression model produced an R^2 score of only 0.002, confirming that rating alone explains less than 0.2% of the variation in install counts.

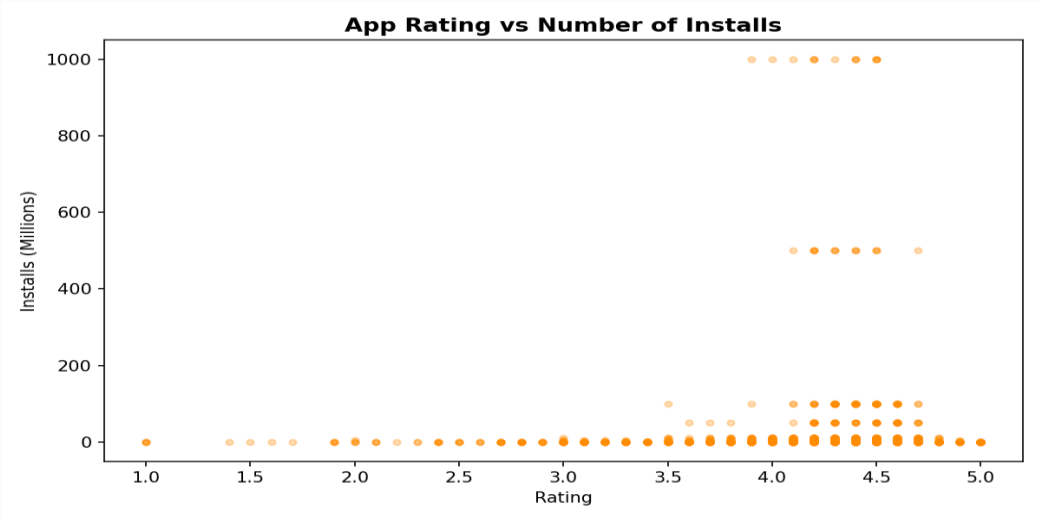

Figure 2. Scatter plot of App Rating vs Number of Installs (sample of 2,000 apps). Each point represents a single app. The plot shows no clear upward trend.

Despite the intuitive assumption that better-rated apps would attract more downloads, the data strongly contradicts this. The scatter plot shows a dense cluster of apps between ratings 4.0 and 4.8 with widely varying install counts — ranging from under 1,000 to over 500 million — suggesting that rating alone does not drive install behaviour. Other variables such as marketing, category popularity, brand recognition, and app discoverability likely play a far greater role. This is an important practical insight for developers: improving ratings alone is unlikely to significantly boost downloads without parallel investment in visibility and marketing.

7.3 Research Question 3 — Which app categories dominate by installs and ratings?

Apps were grouped by Category and the mean number of installs was calculated for each group. The results were sorted in descending order and the top 10 categories were plotted on a bar chart. Additionally, a histogram of ratings was produced to examine the overall distribution of app quality across all categories.

Results: The Communication category led with an average of approximately 78.7 million installs per app, followed by Social (51.3M), Video Players (38.9M), Productivity (37.3M), and Photography (32.0M). The overall app rating distribution was skewed towards higher values, with a mean of 4.19 and 57.5% of all apps rated between 4.0 and 4.5.

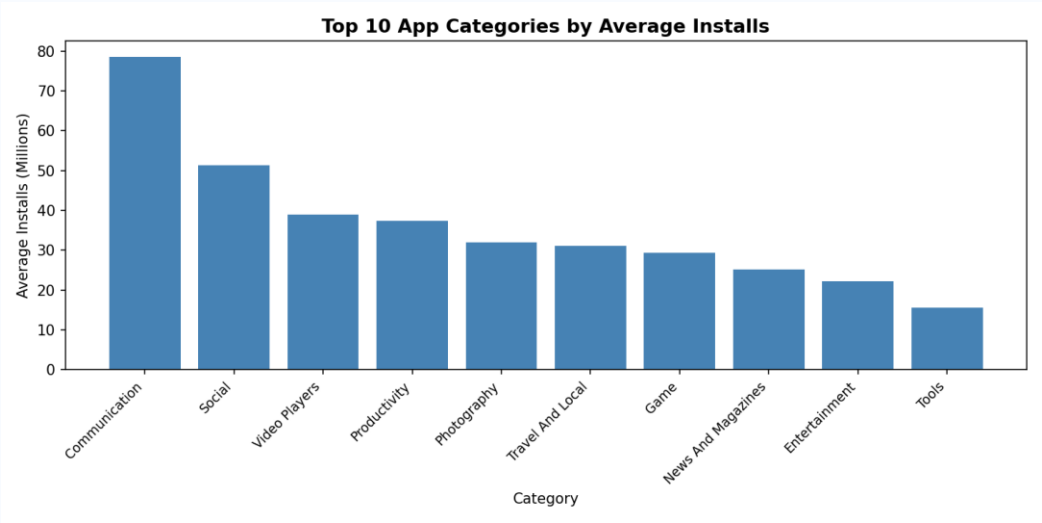

Figure 3. Top 10 App Categories by Average Installs. Communication and Social categories lead by a wide margin due to the presence of globally dominant apps such as WhatsApp and Facebook.

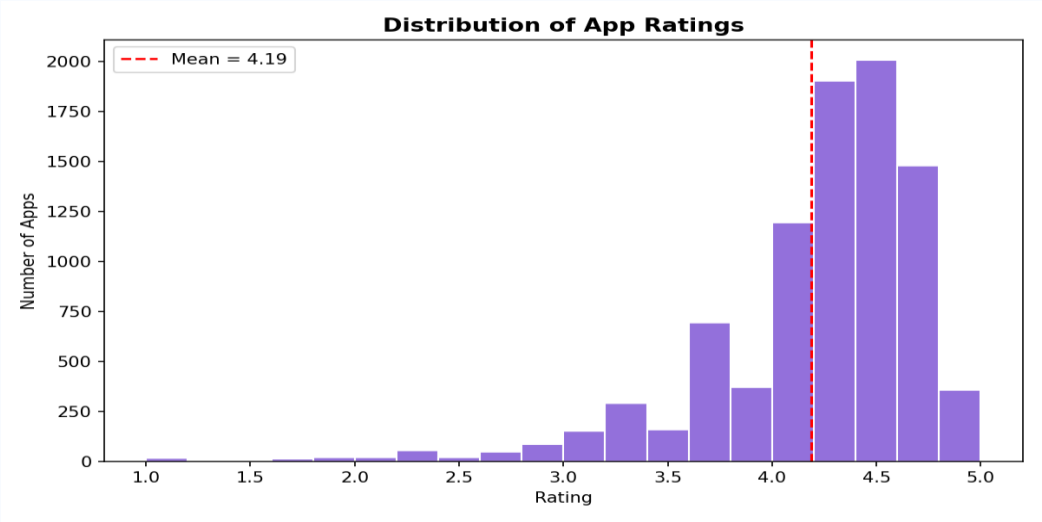

Figure 4. Distribution of App Ratings across all Google Play Store apps. The distribution is left-skewed with a peak between 4.0 and 4.5, and a mean rating of 4.19.

The category analysis reveals that utility-type apps — communication tools, social networks, and video players — dominate install counts because they serve fundamental daily needs for global audiences. These categories benefit from massive network effects (e.g. messaging apps are more valuable when more people use them), which explains the enormous gap between them and niche categories. The rating histogram demonstrates a consistent upward pressure in ratings: most apps cluster between 4.0 and 4.5, suggesting either genuine quality improvement over time or the well-documented survivorship bias, where poorly-rated apps are removed from the store.

8. NumPy-Based Custom Computation

A percentile-based analysis of the Installs distribution was performed using NumPy to understand the spread and skewness of the install data beyond simple averages.

Computation Used	np.percentile() applied to the Installs column at the 10th, 50th (median), and 90th percentiles
Purpose	To characterise the install distribution and identify how concentrated or spread out app popularity is, revealing market inequality
Formula / Logic	p10 = np.percentile(df['Installs'], 10) → 1,000 | p50 = np.percentile(df['Installs'], 50) → 500,000 | p90 = np.percentile(df['Installs'], 90) → 10,000,000
Result / Takeaway	The massive gap between the 10th percentile (1,000 installs) and the 90th percentile (10,000,000 installs) — a 10,000x difference — confirms extreme inequality in the app market. The vast majority of apps attract modest attention while a tiny fraction achieves mass adoption. This insight contextualises the outlier analysis and explains why average install figures can be misleading.

9. Key Findings

The following five findings are directly supported by the quantitative analysis performed in this project:

Finding 1: Free apps receive dramatically more installs than paid apps. The average install count for free apps (17.7 million) is approximately 156 times greater than for paid apps (112,908). This confirms that price is a major barrier to adoption and that the freemium model dominates the Android app ecosystem.

Finding 2: Rating has virtually no predictive power over install counts. A Pearson correlation of 0.05 and a linear regression R^2 of 0.002 demonstrate that knowing an app's rating tells us almost nothing about how many times it will be downloaded. Other factors — brand, marketing, category — are far more influential.

Finding 3: Communication and Social categories dominate the market by installs. Apps in the Communication category average 78.7 million installs, driven by globally dominant platforms like WhatsApp and Google Chrome. This reflects the power of network effects and the essential nature of messaging tools in daily life.

Finding 4: The app market exhibits extreme install inequality. NumPy percentile analysis revealed a 10,000x gap between the 10th percentile (1,000 installs) and the 90th percentile (10,000,000 installs). This means most apps struggle for visibility while a very small number achieve massive scale, resembling a winner-takes-most dynamic.

Finding 5: App ratings are consistently inflated toward the 4.0–4.5 range. With 57.5% of apps rated in this narrow band and a mean of 4.19, ratings do not effectively differentiate app quality. This could reflect survivorship bias (poorly rated apps being removed), developer-encouraged positive reviews, or genuine quality improvement driven by competitive market pressure.

10. Limitations

This project was conducted with methodological care, but several important limitations must be acknowledged:

1. Missing and dropped data: Approximately 1,474 rows with missing ratings were removed. If lower-quality apps are disproportionately missing ratings (e.g. because they are new or unpopular), the cleaned dataset may over-represent well-established, higher-rated apps, introducing selection bias into the findings.

2. Approximate install counts: The dataset records installs in bracket ranges (e.g. '1,000,000+') rather than exact figures. All install-based analysis, including averages, correlations, and percentiles, should be treated as estimates rather than precise measurements.

3. Outdated data: This dataset reflects the Google Play Store at a specific historical point in time. App market dynamics, category composition, and user behaviour may have changed significantly since collection, limiting the generalisability of findings to the present day.

4. Correlation does not imply causation: The weak correlation between rating and installs does not mean ratings are irrelevant to app success. Other unmeasured variables — marketing spend, app store optimisation, social media presence, and external reviews — likely influence both ratings and installs simultaneously.

5. Outlier skew in category averages: A small number of globally dominant apps (e.g. Google, Facebook, WhatsApp) with billions of installs heavily inflate the category averages. The Communication category average of 78.7 million is largely driven by a handful of mega-apps, not representative of a typical communication app.

11. Conclusion

This project successfully applied Pandas, NumPy, and Matplotlib to a real-world dataset of 10,841 Google Play Store apps to answer three meaningful analytical questions about install behaviour, rating influence, and category popularity. After thorough data cleaning, feature engineering, and statistical analysis, the evidence points to clear and actionable conclusions: free apps dramatically outperform paid apps in user adoption; ratings have negligible predictive power over downloads; and communication and social categories dominate the market through network effects and global demand.

The NumPy percentile analysis further revealed a highly unequal distribution of app success, where the gap between the most and least installed apps spans four orders of magnitude. Taken together, these findings suggest that for app developers, pricing strategy and category selection matter far more than optimising ratings alone when targeting mass-market adoption.

Future work could extend this analysis by incorporating time-series data on install growth, using multivariate regression to model installs as a function of rating, reviews, size, and category simultaneously, or applying machine learning classification to predict whether an app falls into a High, Medium, or Low install category based on its attributes.

12. References / Dataset Source

Lava18. (n.d.). Google Play Store Apps [Dataset]. Kaggle. Retrieved from https://www.kaggle.com/datasets/lava18/google-play-store-apps

McKinney, W. (2022). Python for Data Analysis (3rd ed.). O'Reilly Media.

Harris, C. R. et al. (2020). Array programming with NumPy. Nature, 585, 357–362.

Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. Computing in Science & Engineering, 9(3), 90–95.

Final Submission Checklist

☑ Dataset selected with 10,841 rows and 13 columns from Kaggle.

☑ Pandas, NumPy, and Matplotlib used meaningfully throughout the project.

☑ 6 cleaning and preprocessing steps performed with full justification.

☑ 3 engineered features created (Revenue, Installs_Category, Year).

☑ 6+ meaningful analysis operations performed (subgroup comparison, correlation, outlier, ranking, distribution, percentile).

☑ 2 subgroup comparisons included (Free vs Paid; Category-level grouping).

☑ 1 relationship analysis (Rating vs Installs correlation + regression).

☑ 1 outlier analysis (top 1% apps by installs; 49 extreme-outlier apps identified).

☑ 4 Matplotlib charts produced with titles, labelled axes, and interpretations.

☑ 5 evidence-based findings written and supported by data.

☑ Limitations section included discussing bias, data quality, and causation caveats.

☑ Report complete and ready to submit.